

Bash Troubleshooting Commands Guide

Essential Bash Commands for Troubleshooting

System Information & Diagnostics

Command	Description	Example
<code>uname -a</code>	System information	<code>uname -a</code> (kernel version, architecture, hostname)
<code>hostname</code>	Display hostname	<code>hostname -f</code> (fully qualified domain name)
<code>uptime</code>	System uptime and load	<code>uptime -p</code> (pretty format)
<code>whoami</code>	Current user	<code>whoami</code>
<code>id</code>	User ID and group info	<code>id username</code>
<code>date</code>	Date and time	<code>date -R</code> (RFC 2822 format)
<code>env</code>	Environment variables	<code>env grep -i path</code>
<code>printenv</code>	Print environment	<code>printenv PATH</code>
<code>df -h</code>	Disk usage in human-readable	<code>df -h /dev/sda1</code>
<code>du -sh</code>	Directory size	<code>du -sh /var/log/* sort -hr</code>
<code>free -m</code>	Memory usage in MB	<code>free -m && free -h</code>
<code>top</code>	Real-time process monitoring	<code>top -u username</code> (filter by user)
<code>htop</code>	Enhanced process monitoring	<code>htop -p PID</code> (monitor specific process)
<code>ps aux</code>	All running processes	<code>ps auxf</code> (tree view)
<code>pstree -p</code>	Process tree with PIDs	<code>pstree -p grep -i ssh</code>
<code>cat /proc/cpuinfo</code>	CPU information	<code>cat /proc/cpuinfo grep "model name"</code>
<code>cat /proc/meminfo</code>	Memory information	<code>cat /proc/meminfo grep -E "^(MemTotal MemFree SwapTotal SwapFree) "</code>

Command	Description	Example
lscpu	CPU architecture details	lscpu
lsblk	Block device information	lsblk -f (with filesystem info)
fdisk -l	Disk partition table	sudo fdisk -l

Network Troubleshooting

Command	Description	Example
ping	ICMP echo test	ping -c 4 google.com
tracert	Route path diagnosis	tracert -n 8.8.8.8 (no DNS resolution)
tracert	Path MTU discovery	tracert -n google.com
mtr	Network diagnostics (my traceroute)	mtr -r -c 10 google.com (report mode)
nslookup	DNS query	nslookup google.com
dig	DNS lookup tool	dig google.com A +short
host	DNS lookup utility	host -t MX google.com
ip	Network interface and routing	ip addr show and ip route show
ifconfig	Network interface config (legacy)	ifconfig -a
ss	Socket statistics (replaces netstat)	ss -tulpn (TCP/UDP listening ports)
netstat	Network statistics	netstat -tulpn grep LISTEN
lsof -i	Network connections by process	sudo lsof -i :80 (port 80)
curl	HTTP requests	curl -v -I https://example.com (verbose headers)
wget	HTTP/HTTPS/FTP download	wget --spider -S https://example.com (check headers only)
nc (netcat)	TCP/UDP connection testing	nc -zv google.com 80 (port 80)
telnet	Connection test	telnet google.com 443
arp	ARP cache	arp -a
iptables	Firewall rules	sudo iptables -L -n -v
tcpdump	Packet capture	sudo tcpdump -i eth0 -n port 443

File System & Permissions

Command	Description	Example
<code>ls -la</code>	Detailed file listing	<code>ls -la /etc sort -k5 -hr</code> (sort by size)
<code>find</code>	File search	<code>find /var/log -name "*.log" -mtime -1</code> (modified in last 24h)
<code>locate</code>	Fast file search (updatedb)	<code>locate -i "nginx.conf"</code>
<code>which</code>	Command location	<code>which python3</code>
<code>whereis</code>	Binary/source/man location	<code>whereis nginx</code>
<code>type</code>	Command type	<code>type -a ls</code> (all locations)
<code>file</code>	File type	<code>file /bin/ls</code>
<code>stat</code>	File metadata	<code>stat /etc/passwd</code>
<code>chmod</code>	Change permissions	<code>chmod 644 file.txt</code>
<code>chown</code>	Change ownership	<code>chown user:group file.txt</code>
<code>umask</code>	Default permissions	<code>umask 022</code>
<code>acl</code>	Access Control Lists	<code>getfacl /path/to/file</code>
<code>ln -l</code>	List symbolic links	<code>ls -la grep "^l"</code>
<code>mount</code>	Mounted filesystems	<code>mount -t ext4</code>
<code>umount</code>	Unmount filesystem	<code>umount /mnt/usb</code>
<code>df -i</code>	Inode usage	<code>df -i /dev/sda1</code>

Process Management & Performance

Command	Description	Example
<code>kill</code>	Terminate process	<code>kill -9 PID</code> (force kill)
<code>killall</code>	Kill by name	<code>killall -9 process_name</code>
<code>pkill</code>	Pattern-based kill	<code>pkill -f "python.*script"</code>
<code>nice</code>	Set priority	<code>nice -n 10 heavy_process</code>
<code>renice</code>	Change running process priority	<code>renice -n 15 -p PID</code>
<code>nohup</code>	Process immune to hangups	<code>nohup command &</code>
<code>disown</code>	Remove from shell job table	<code>disown %1</code>
<code>jobs</code>	List background jobs	<code>jobs -l</code> (with PIDs)
<code>fg</code>	Bring to foreground	<code>fg %1</code>
<code>bg</code>	Send to background	<code>bg %1</code>
<code>time</code>	Command timing	<code>time heavy_script.sh</code>
<code>strace</code>	System call tracing	<code>strace -p PID -e trace=file</code>
<code>ltrace</code>	Library call tracing	<code>ltrace -p PID</code>
<code>vmstat</code>	Virtual memory statistics	<code>vmstat -s</code>
<code>iostat</code>	I/O statistics	<code>iostat -x 2</code> (every 2 seconds)

Command	Description	Example
mpstat	CPU statistics	mpstat -P ALL 2
sar	System activity reporter	sar -u 2 10 (CPU every 2s, 10 times)

Log Analysis & Monitoring

Command	Description	Example
tail -f	Follow log file	tail -f /var/log/syslog
head	First lines	head -n 20 /var/log/auth.log
tail	Last lines	tail -n 100 /var/log/syslog
grep	Pattern matching	grep -i "error" /var/log/syslog
less	Interactive file viewing	less +F /var/log/syslog (follow mode)
more	File viewer	more /var/log/nginx/access.log
awk	Text processing	awk '{print \$1, \$4}' access.log
sed	Stream editor	sed -n '/error/p' logfile
journalctl	Systemd journal	journalctl -u nginx -f
dmesg	Kernel ring buffer	dmesg -T tail -20 (human timestamps)
systemd-analyze	Boot performance	systemd-analyze blame

User & Group Management

Command	Description	Example
who	Who is logged in	who -u (with idle time)
w	Who is logged in and what they're doing	w
last	Login history	last -n 10
lastlog	Last login report	lastlog -u username
groups	Group memberships	groups username
adduser	Add user	sudo adduser username
deluser	Delete user	sudo deluser --remove-home username
usermod	Modify user	sudo usermod -aG sudo username
passwd	Change password	passwd username
chage	Password aging info	chage -l username
su	Switch user	su - username
sudo	Execute as superuser	sudo -u user command
visudo	Edit sudoers file	sudo visudo

Package Management (Debian/Ubuntu)

Command	Description	Example
<code>apt update</code>	Update package lists	<code>sudo apt update</code>
<code>apt upgrade</code>	Upgrade packages	<code>sudo apt upgrade -y</code>
<code>apt search</code>	Search packages	<code>apt search nginx</code>
<code>apt show</code>	Package details	<code>apt show nginx</code>
<code>apt list --installed</code>	List installed packages	<code>apt list --installed grep nginx</code>
<code>dpkg -l</code>	List packages (legacy)	<code>dpkg -l grep nginx</code>
<code>dpkg -S</code>	File ownership	<code>dpkg -S /etc/nginx/nginx.conf</code>
<code>apt-cache depends</code>	Package dependencies	<code>apt-cache depends nginx</code>
<code>aptitude</code>	Interactive package manager	<code>sudo aptitude</code>

Package Management (RHEL/CentOS/Fedora)

Command	Description	Example
<code>yum update</code>	Update packages	<code>sudo yum update -y</code>
<code>yum install</code>	Install package	<code>sudo yum install nginx</code>
<code>yum search</code>	Search packages	<code>yum search nginx</code>
<code>yum info</code>	Package information	<code>yum info nginx</code>
<code>yum list installed</code>	List installed	<code>yum list installed grep nginx</code>
<code>rpm -qa</code>	List all installed RPMs	<code>rpm -qa grep nginx</code>
<code>rpm -qf</code>	File ownership	<code>rpm -qf /etc/nginx/nginx.conf</code>
<code>dnf</code>	DNF (newer Fedora)	<code>sudo dnf install nginx</code>

System Services & Boot

Command	Description	Example
<code>systemctl status</code>	Service status	<code>systemctl status nginx</code>
<code>systemctl start</code>	Start service	<code>sudo systemctl start nginx</code>
<code>systemctl stop</code>	Stop service	<code>sudo systemctl stop nginx</code>
<code>systemctl restart</code>	Restart service	<code>sudo systemctl restart nginx</code>
<code>systemctl reload</code>	Reload config	<code>sudo systemctl reload nginx</code>
<code>systemctl enable</code>	Enable on boot	<code>sudo systemctl enable nginx</code>
<code>systemctl disable</code>	Disable on boot	<code>sudo systemctl disable nginx</code>
<code>systemctl list-units</code>	List all services	<code>systemctl list-units --type=service</code>
<code>systemctl list-unit-files</code>	All unit files	<code>systemctl list-unit-files grep enabled</code>
<code>systemctl is-active</code>	Check if active	<code>systemctl is-active nginx</code>
<code>systemctl is-enabled</code>	Check if enabled	<code>systemctl is-enabled nginx</code>

Command	Description	Example
service	Legacy service command	sudo service nginx status

SSH & Remote Access Troubleshooting

Command	Description	Example
ssh -v	Verbose SSH connection	ssh -v user@host
ssh -vvv	Maximum verbosity	ssh -vvv user@host
ssh-keygen	Generate SSH keys	ssh-keygen -t rsa -b 4096
ssh-copy-id	Copy public key	ssh-copy-id user@host
scp -v	Secure copy with verbose	scp -v file user@host:/path/
rsync -av	Sync with progress	rsync -avz --progress src/ user@host:/dest/
ssh-add	Add key to agent	ssh-add ~/.ssh/id_rsa
ssh-agent	SSH agent	eval \$(ssh-agent -s)

Common Troubleshooting Scenarios

Case: CPU High/Load Average

```
bash
# Check CPU usage
top -b -n 1 | head -15
ps aux --sort=-%cpu | head -10
mpstat -P ALL 1 5

# Find process using CPU
ps -eo pid,ppid,cmd,%cpu --sort=-%cpu | head -10
htop -u username
```

Case: Memory High Usage

```
bash
# Memory usage overview
free -mh
vmstat -s | grep -E "memory|swap"
ps aux --sort=-%mem | head -10

# Find top memory consumers
ps -eo pid,ppid,cmd,%mem --sort=-%mem | head -10

# Check memory by process
smem -r -p | head -10
```

Case: Disk Space Full

```
bash
```

```
# Find large directories
du -sh /* 2>/dev/null | sort -hr | head -10
du -sh /home/* 2>/dev/null | sort -hr | head -10

# Find large files
find / -type f -size +100M -exec ls -lh {} \; 2>/dev/null
find / -type f -exec du -h {} + 2>/dev/null | sort -hr | head -10

# Check inode usage
df -i

# Clear logs
sudo journalctl --vacuum-size=100M
sudo find /var/log -type f -name "*.log" -size +100M -exec truncate -s 0 {} \;
```

Case: Network Connection Issues

```
bash
# Multi-layer network test
ping -c 4 google.com
traceroute -n google.com
dig google.com A +short
telnet google.com 80
curl -v https://google.com

# Check local network config
ip addr show
ip route show
ss -tulpn

# Check connectivity to common services
nc -zv 8.8.8.8 53
nc -zv 1.1.1.1 443
```

Case: Service Won't Start

```
bash
# Check service status
systemctl status nginx --no-pager -l

# Check logs
journalctl -u nginx -xe -n 50
tail -f /var/log/nginx/error.log

# Check config syntax
nginx -t

# Check port conflicts
ss -tulpn | grep :80
lsof -i :80

# Check permissions
ls -la /etc/nginx/nginx.conf
stat /var/log/nginx
```

Quick One-Liners for Troubleshooting

bash

```
# Check system load and processes
uptime && ps aux --sort=-%cpu | head -5

# Check disk usage summary
df -h && echo "---" && du -sh /home/* 2>/dev/null | sort -hr | head -5

# Check memory and top processes
free -h && echo "---" && ps aux --sort=-%mem | head -5

# Check network stats
ss -s && echo "---" && iptables -L -n -v | head -20

# Check user sessions
who -u && echo "---" && lastlog -t 7

# Check recent system events
journalctl -xe --since "1 hour ago" | tail -20

# Check for failed login attempts
grep "Failed password" /var/log/auth.log | tail -10
sudo journalctl -u ssh -n 50 | grep -i "fail"

# Check open ports
ss -tulpn | grep LISTEN

# Check DNS resolution
dig google.com +short && nslookup google.com | grep Server

# Check date/time synchronization
timedatectl status
```

Essential Bash Shortcuts

Shortcut	Description
Ctrl + C	Interrupt/stop current command
Ctrl + Z	Suspend current process
Ctrl + D	End of file/exit shell
Ctrl + L	Clear screen
Ctrl + R	Reverse search history
Ctrl + A	Go to beginning of line
Ctrl + E	Go to end of line
!!	Repeat last command
!\$	Last argument of previous command
!!:2	Second argument of previous command
Ctrl + Arrow	Jump by words
Tab	Auto-completion

Shortcut	Description
Ctrl + Shift + C	Copy (terminal)
Ctrl + Shift + V	Paste (terminal)

Useful Environment Variables

Variable	Description
\$HOME	Home directory
\$PATH	Command search path
\$SHELL	Current shell
\$USER	Current username
\$PWD	Current directory
\$OLDPWD	Previous directory
\$?	Exit status of last command
\$PS1	Prompt string
\$LANG	Language setting
\$TERM	Terminal type
\$EDITOR	Default editor

Command Chaining & Redirection

Operator	Description	Example
;	Execute sequentially	command1 ; command2
&&	Execute if previous succeeded	mkdir dir && cd dir
	Execute if previous failed	command1 echo "Failed"
	Pipe output to next command	ps aux grep nginx
>	Redirect output (overwrite)	echo "text" > file.log
>>	Redirect output (append)	echo "text" >> file.log
2>	Redirect stderr	command 2> error.log
2>&1	Redirect stderr to stdout	command > all.log 2>&1
<	Redirect input	sort < file.txt